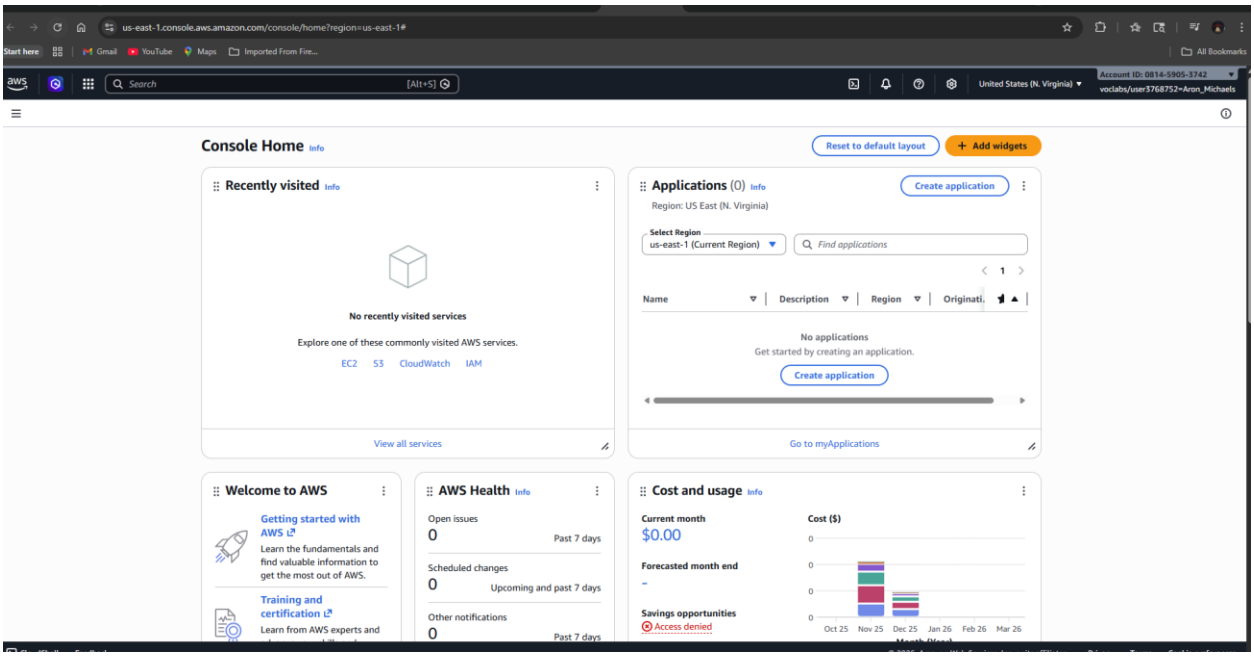
Week 1 Project Template Aron Michaels

CMIT 495 Current Trends and Projects in Computer Networks and Security

Week 1 – Virtualization

1. Access **your AWS** Learner Lab Virtual Environment **and take a screen capture of the AWS Console (Dashboard) and embed it below.**



2. **Provide a detailed overview of the steps required to provision (create) and launch an AWS Ubuntu-based server virtual instance. The steps may be listed in bullet points or complete sentences. Use as much space as required.**

To provision and launch the AWS Ubuntu-based server virtual instance as a proof of concept for Caduceus Inc.'s expansion of shared computing resources (Scenario Overview, Page 1), I followed these steps in the EC2 launch wizard, focusing on security and virtualization best practices (NIST SP 800-125, Page 1 on guide to security).

Step 1: Start the Launch Wizard

- What I did: Navigated to the EC2 Dashboard and clicked 'Launch instance'.
- Why: Initiates the provisioning to create a virtual machine for cloud scalability (Learning Resource, Page 1 on virtualization allowing multiple OS on one host).

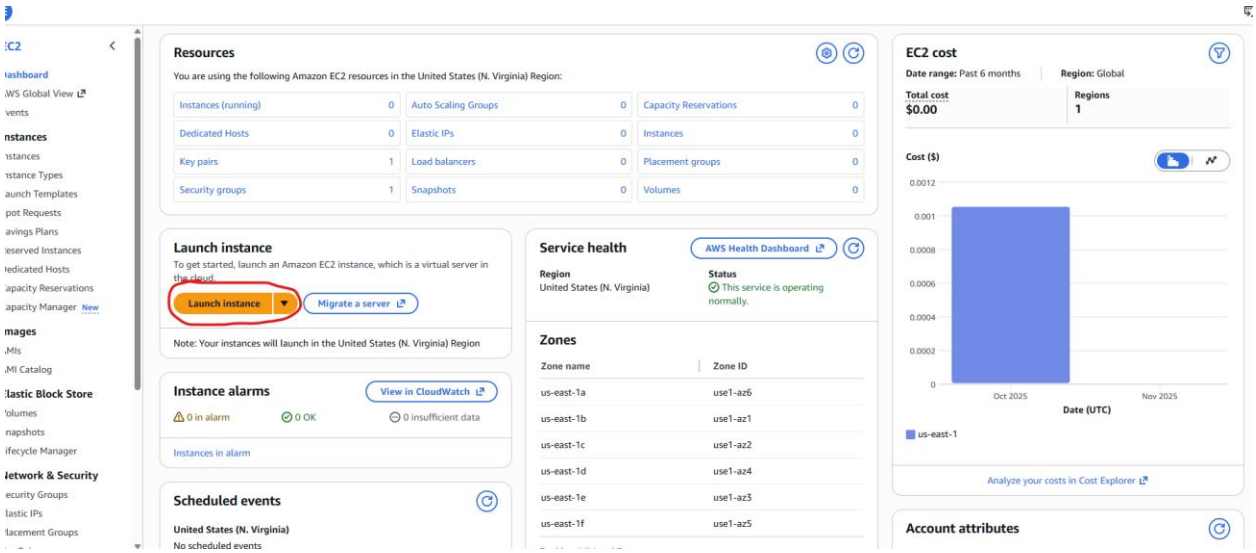


Figure 2c: AMI Selection

Step 2: Name the Instance

- What I did: Entered 'AM-Ubuntu-VM' as name and added tag (Key: 'Project', Value: 'CMIT495').
- Why: Helps organize and track VMs in a cybersecurity setup (NIST SP 800-125, Page 4 on security design).

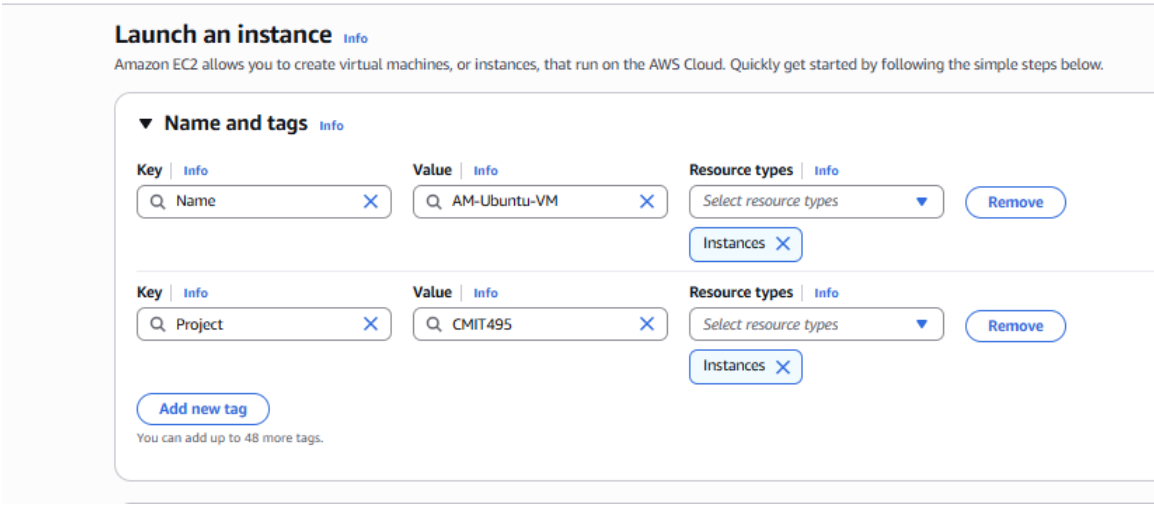


Figure 2b: Instance Name and Tags

Step 3: Select OS Image (AMI)

- What I did: Browsed AMIs, searched 'Ubuntu Server', selected Ubuntu Server 24.04 LTS (64-bit x86, Canonical, free tier).
- Why: Provides the guest OS for VM isolation (Learning Resource, Page 2 on guest OS running like physical).

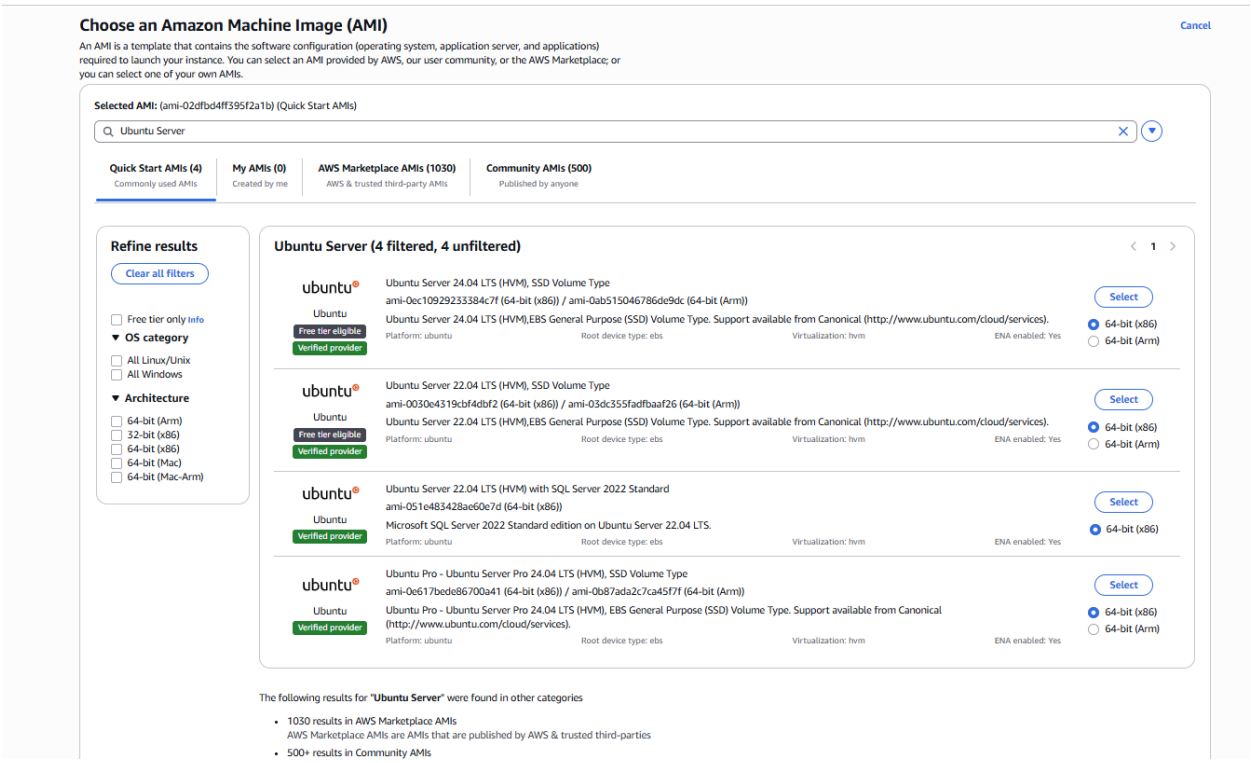


Figure 2c: AMI Selection

Step 4: Choose Instance Type

- What I did: Selected 't3.micro'.
- Why: Low-cost testing (Learning Resource, Page 1 on hardware).

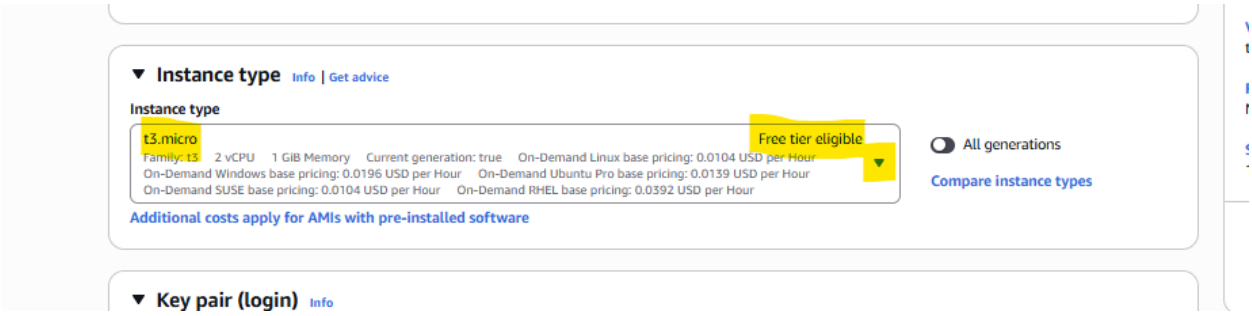


Figure 2d: Instance Type

Step 5: Create Key Pair

- What I did: Created 'AM-ubuntu-key' (.pem), downloaded, set permissions.
- Why: Secure SSH (NIST SP 800-125A, Page 1

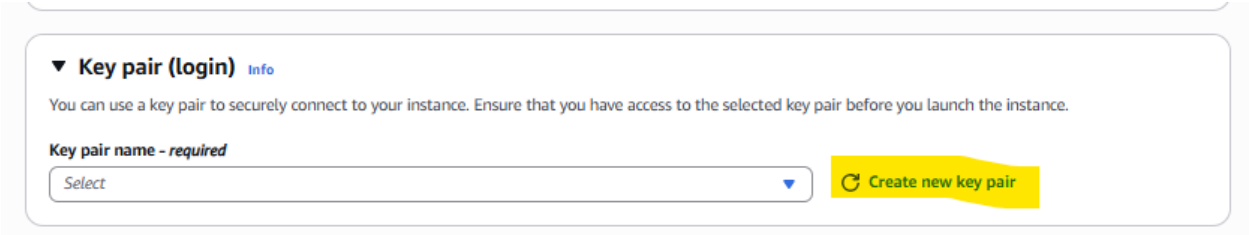


Figure 2e: 1 Key Pair

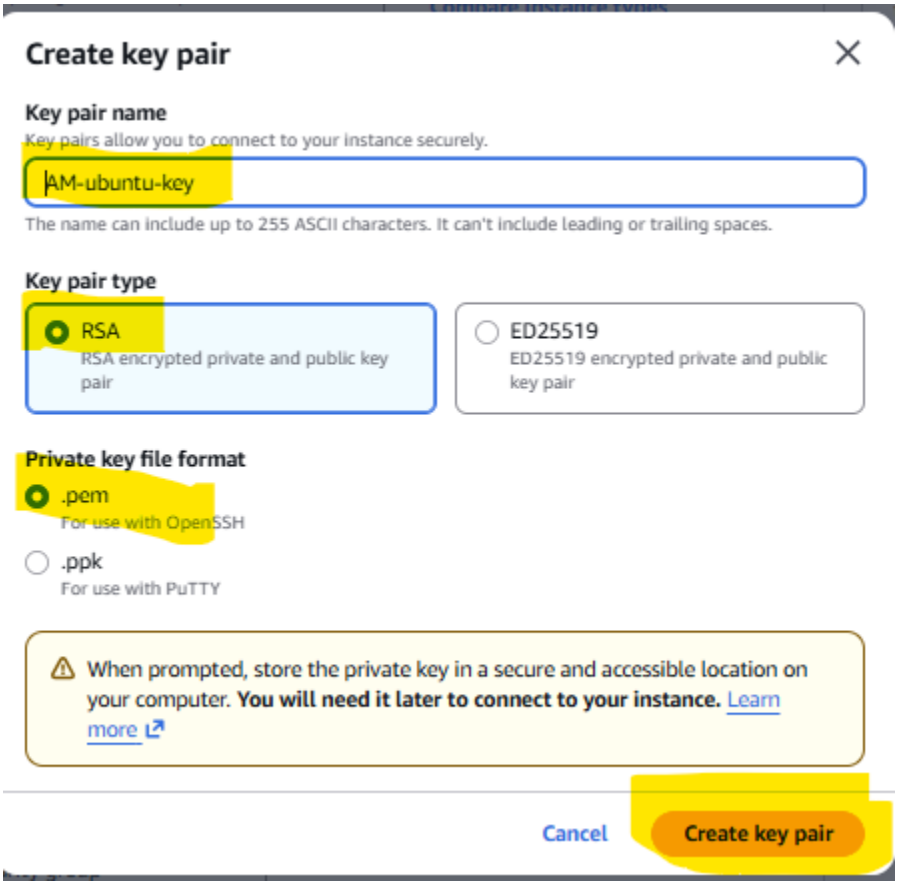


Figure 2e:2 Key Pair

Step 6: Configure Network Settings

- What I did: Kept the default VPC and subnet, then created a new security group named 'am-ssh-security' with an SSH rule (Port 22, Source: My IP only fixed from default Anywhere for security).
- Why: This sets up virtual networking and a firewall for isolation, restricting access to prevent threats (NIST SP 800-125B on secure VM protection; Learning Resource on virtual firewalls and host-only networking).

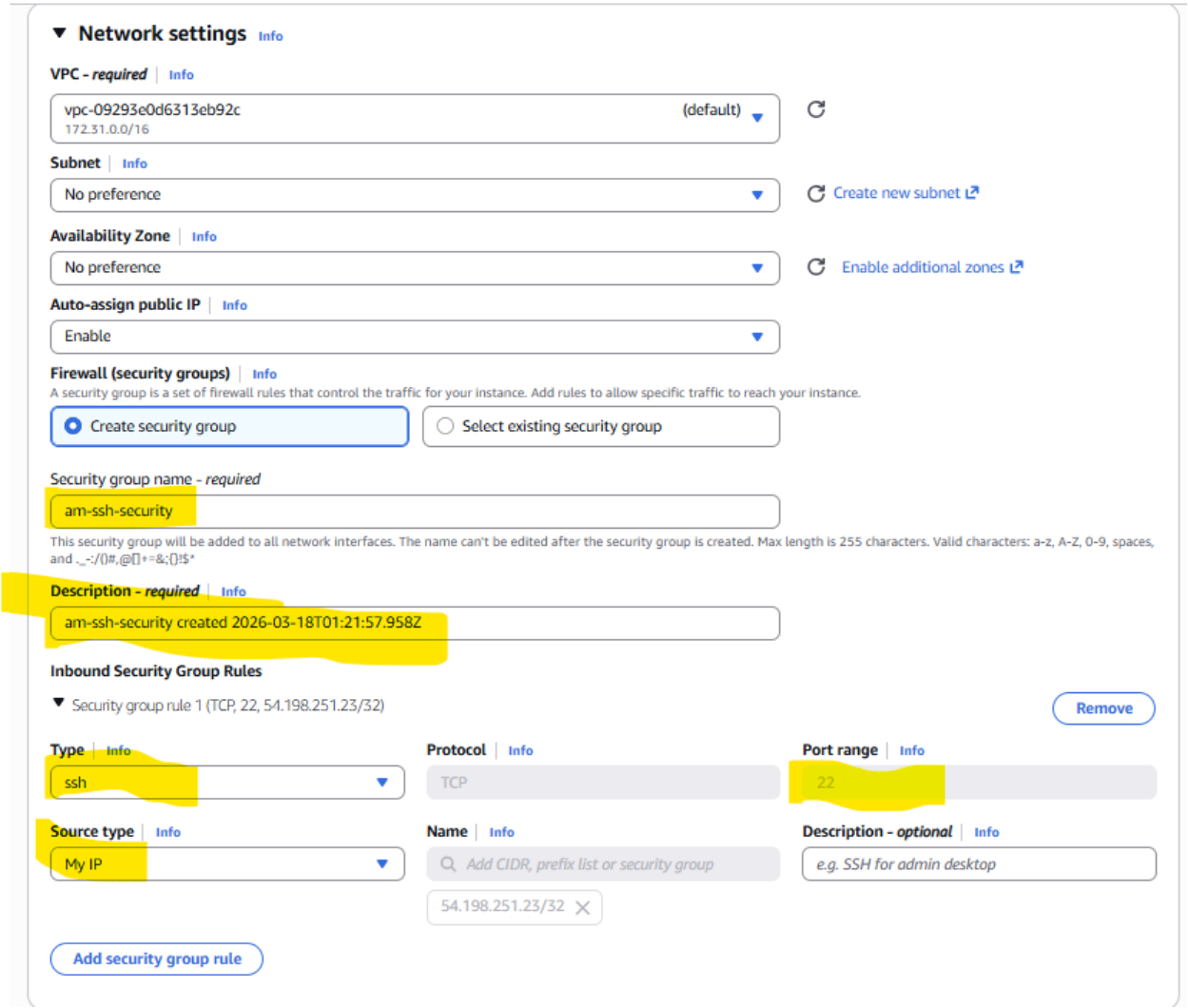


Figure 2f: Network and Security Group

Step 7: Configure Storage

- What I did: Retained the default configuration of 8 GiB gp3 root volume.
- Why: This provides basic virtual disk space for the OS while keeping costs minimal in the lab environment (Learner Lab instructions on preserving budget).

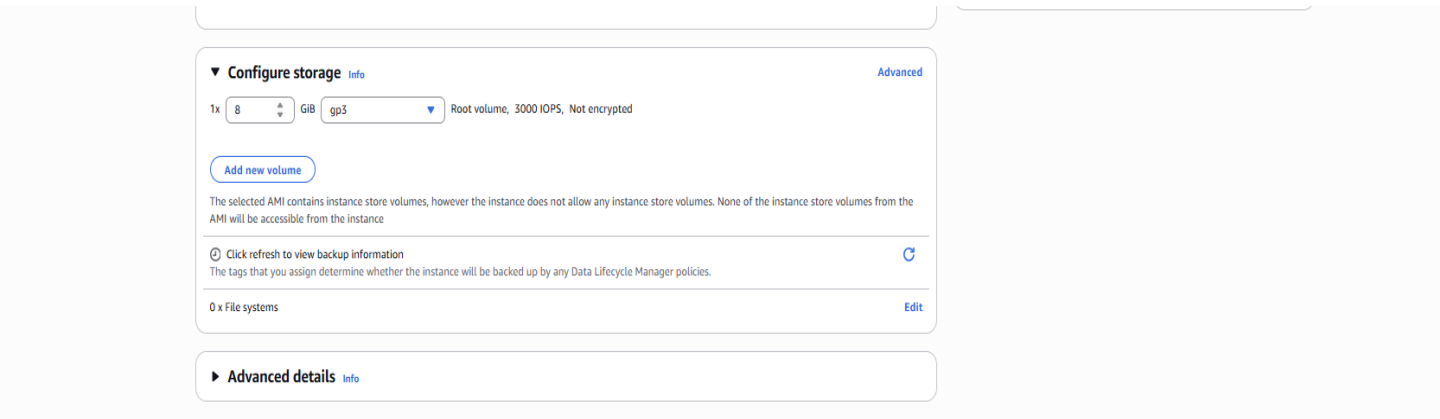


Figure 2g: Storage Configuration

Step 8: Review of Advanced Details

- What I did: Left the advanced details at their default settings, as no custom scripts were needed.
- Why: Defaults are sufficient for a simple PoC, avoiding unnecessary complexity in virtualization setup (Learning Resource on basic virtual machines).

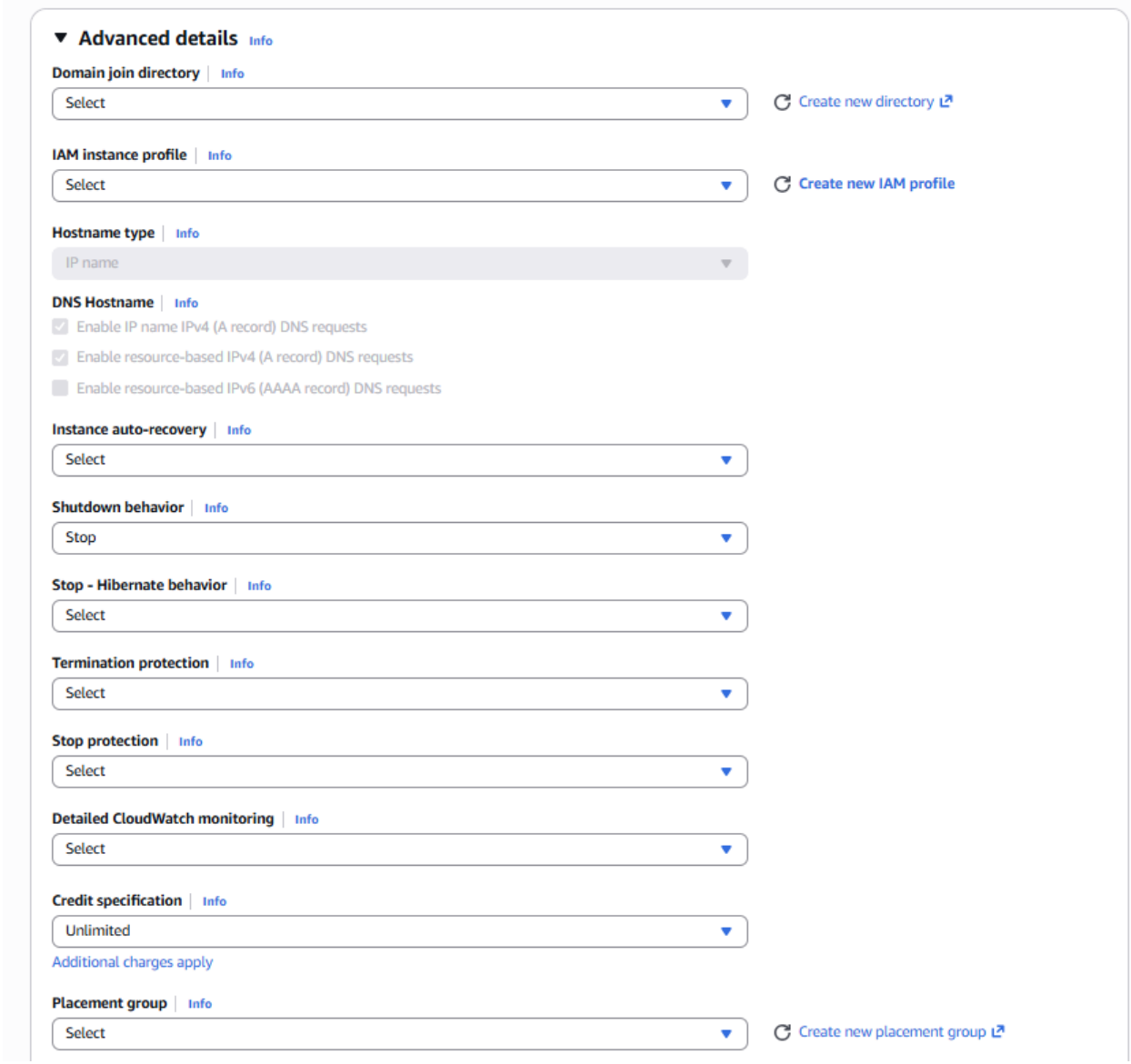


Figure 2h: Advanced Details (Defaults).

Step 9: Launch and Verify the Instance

- What I did: Reviewed the summary pane to confirm settings (e.g., 1 instance, Ubuntu 24.04 AMI, t3.micro, am-ubuntu-key, am-ssh-security group), then clicked 'Launch instance'. After success, monitored the instance details until it showed 'Running' status with 2/2 status checks passed.

- Why: This completes the provisioning, creating a functional virtual server ready for configuration and testing (Scenario Overview on expanding resources; NIST SP 800-125 on full virtualization).

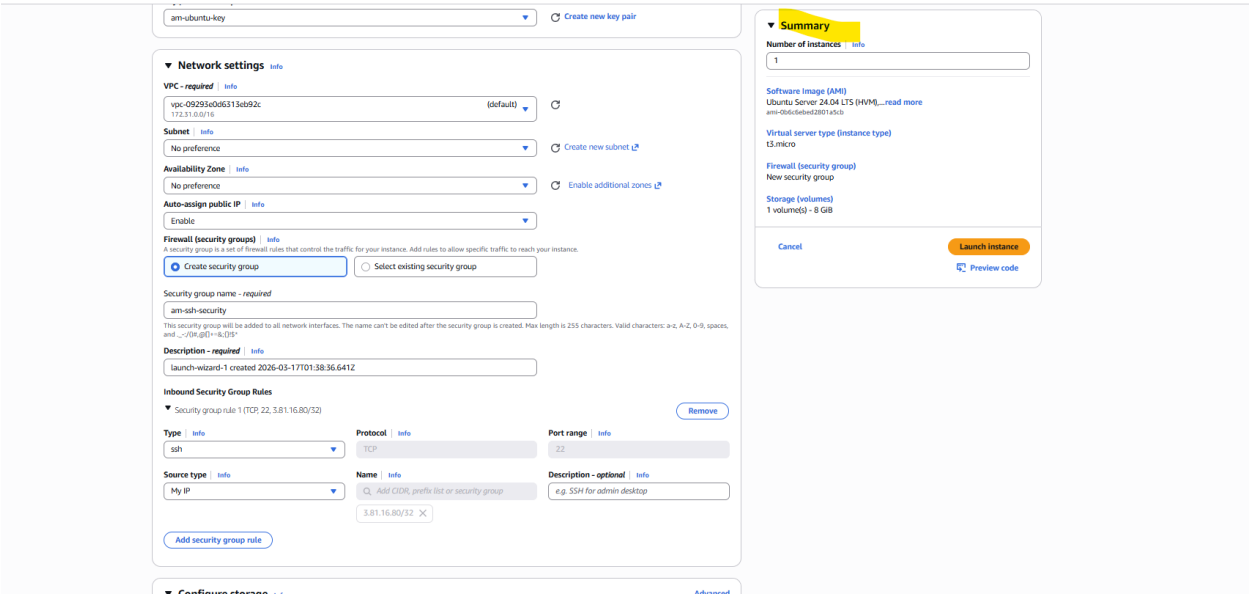
- 

Figure 2i: Launch Summary

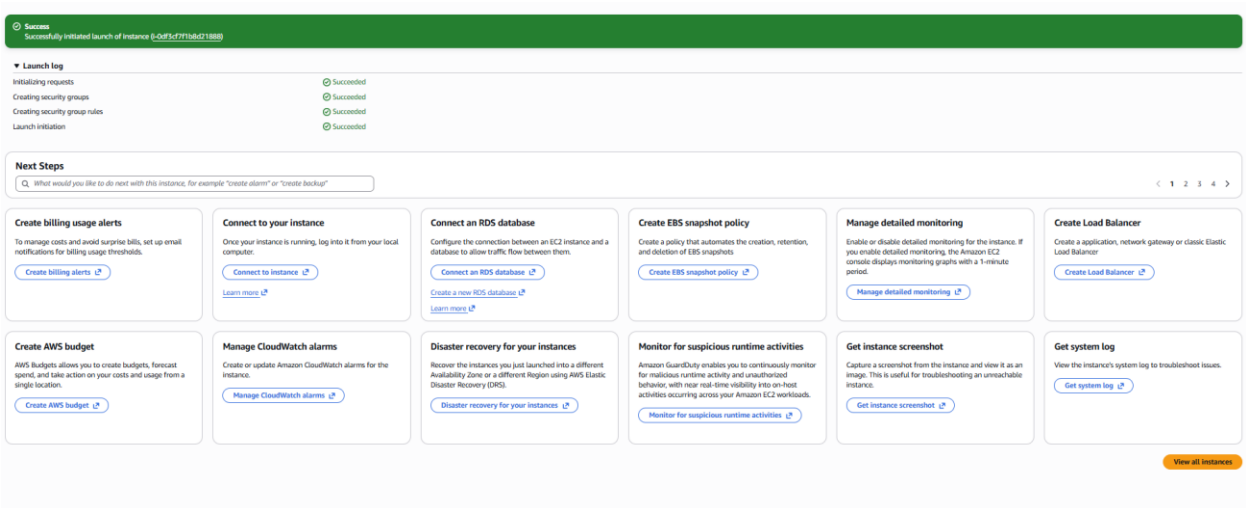
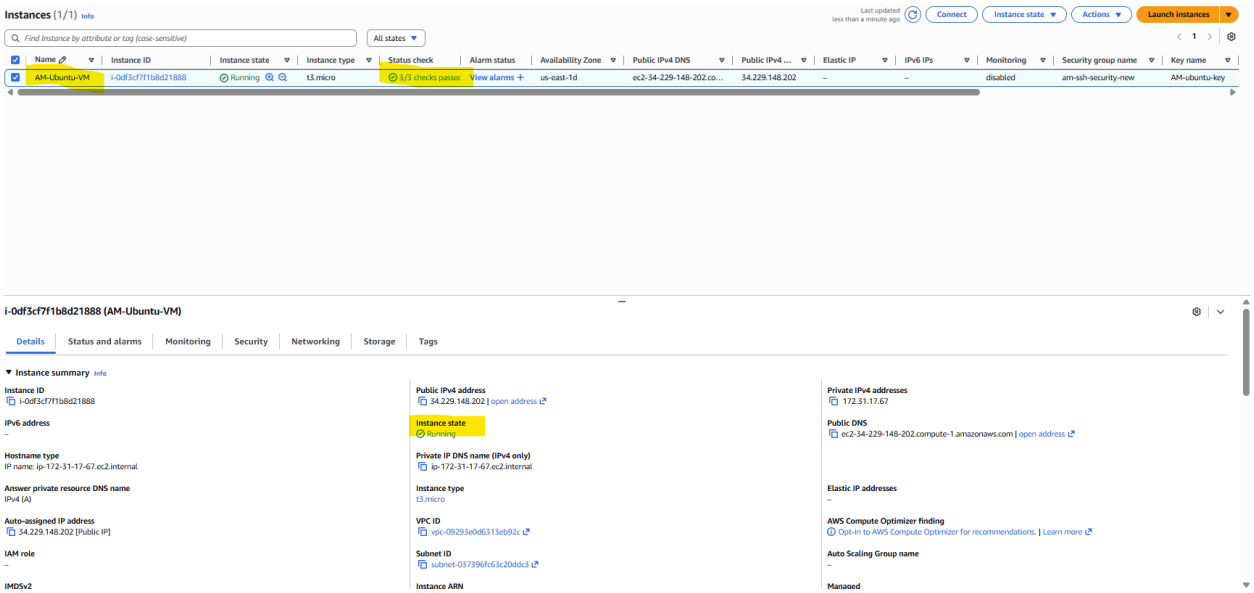


Figure 2j: Launch Success

Figure 2k: Instance Running Status



These steps took about 10 minutes and successfully launched the VM, with a focus on security configurations to meet organizational objectives like protecting client data (NIST SP 800-125).

3. **What are the benefits of virtualization in a cloud environment? Discuss a minimum of three benefits in detail.**

- Benefit 1: Isolation - Virtualization allows separation between VMs, so if one is compromised, it doesn't affect others (Learning Resource on isolation between virtual machines). In detail, this is crucial for Caduceus Inc.'s telehealth, as it protects sensitive health data by containing threats, like walls between rooms, and enables safe testing of software without risking the host system.

- Benefit 2: Scalability - You can easily add or adjust resources like CPU/memory without new hardware (NIST SP 800-125 on advanced network design). In detail, for Caduceus's business surge, this means quickly launching more VMs to handle

increased online presence, reducing costs and time compared to physical servers, with cloud auto-scaling for demand.

- Benefit 3: Portability and Encapsulation - VMs are self-contained files that can be moved, backed up, or restored easily (Learning Resource on encapsulation). In detail, this benefits Caduceus by allowing quick migration between hosts for maintenance or disaster recovery, ensuring availability (CIA triad), and encrypting VM files to maintain confidentiality during transfers

4. **Based on your experience, what was the most challenging aspect of provisioning and launching the AWS Ubuntu-based server instance in the AWS virtual environment?**

As a student with no prior IT experience, the most challenging aspect was configuring the security group in the network settings step. I had to ensure the SSH rule was restricted to my IP to prevent unauthorized access, but understanding how "My IP" auto-detected my location and why it was important for firewall protection took some time (Learner Lab instructions on SSH access; NIST SP 800-125B on secure virtual network configuration). This highlighted the importance of virtual firewalls in isolation (Learning Resource), but once I got it, it made the VM more secure for the PoC.

5. **Configure your local host computer to use an SSH client application to interact with the newly created and running AWS Ubuntu virtual instance. If using a Windows-based local computer, read over the AWS document, "Connect to your Linux instance from Windows using PuTTY", download and install PuTTY[1] or use Windows 10 built-in OpenSSH client.**

[1] *Link: https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html*

Configure your local host computer to use an SSH client application to interact with the newly created and running AWS Ubuntu virtual instance. If using a Windows-based local computer, read over the AWS document, "Connect to your Linux instance from Windows using PuTTY", download and install PuTTY or use Windows 10 built-in OpenSSH client. On my Windows-based local computer, I used PuTTY (already installed). I converted my .pem key to .ppk using PuTTYgen, configured the session with my instance's Public IP and the .ppk file, and connected successfully to the running AWS Ubuntu virtual instance (Learner Lab instructions.pdf, Pages 19-23 on SSH access from Windows).

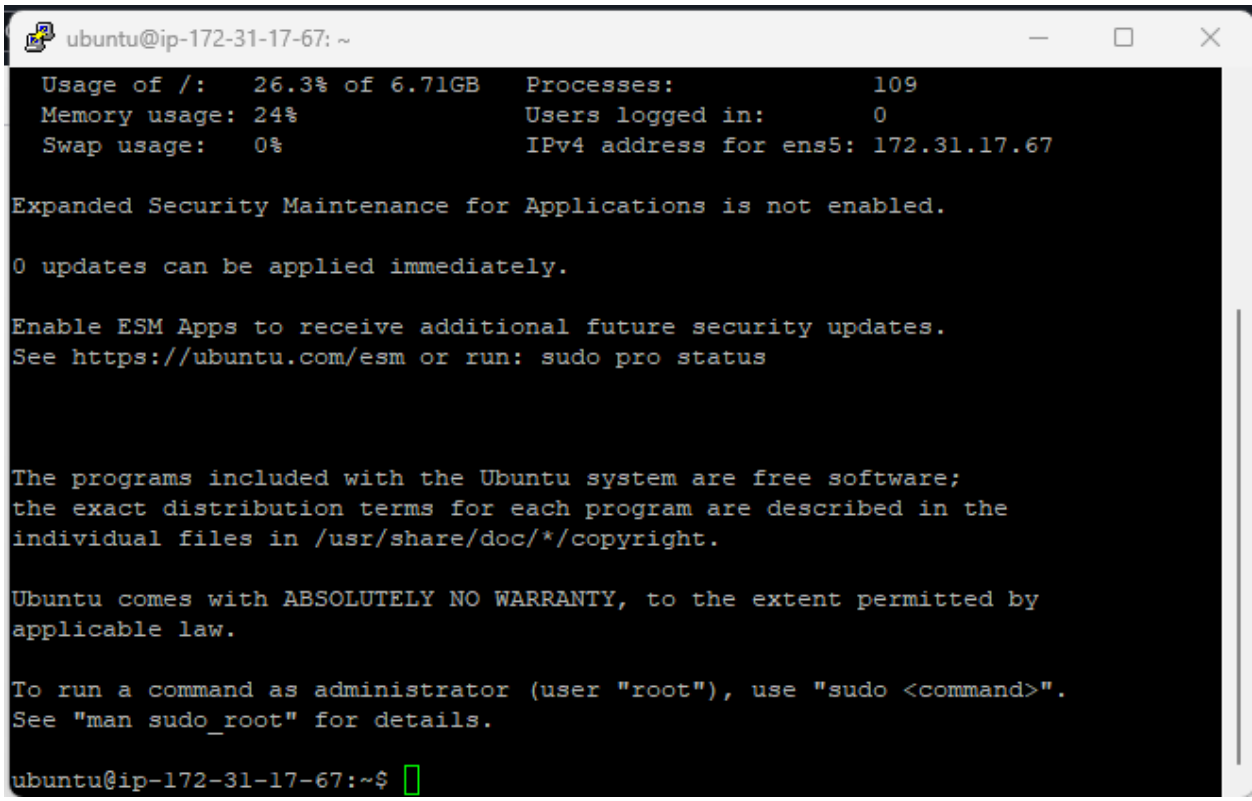


Figure 5: PuTTY SSH Connection Success

6. **From the local computer establish an SSH connection to the AWS Ubuntu Server virtual instance, login, and update the Ubuntu Server using *sudo apt-get update* and *sudo apt-get upgrade*. Perform a screen capture after having each command successfully run.**

I reconnected using PuTTY and ran the update and upgrade commands successfully.

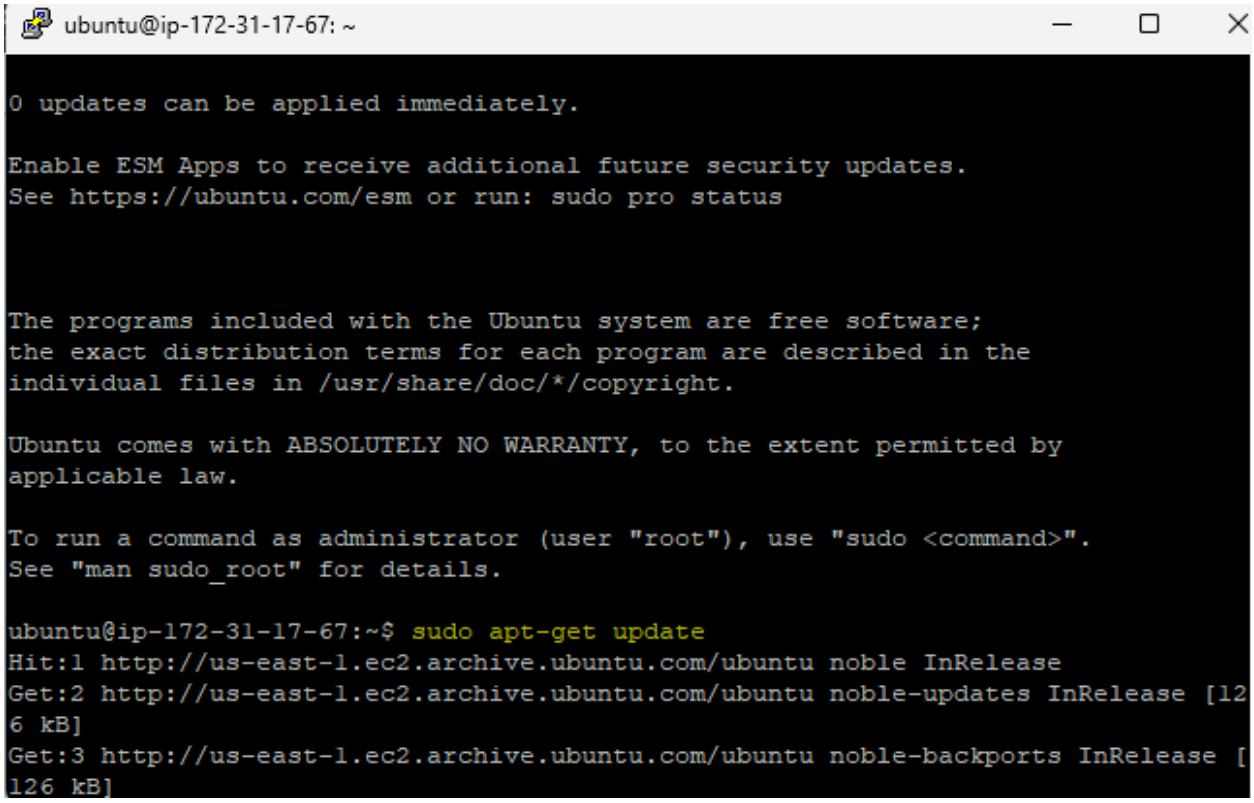


Figure 4 a: sudo apt-get update

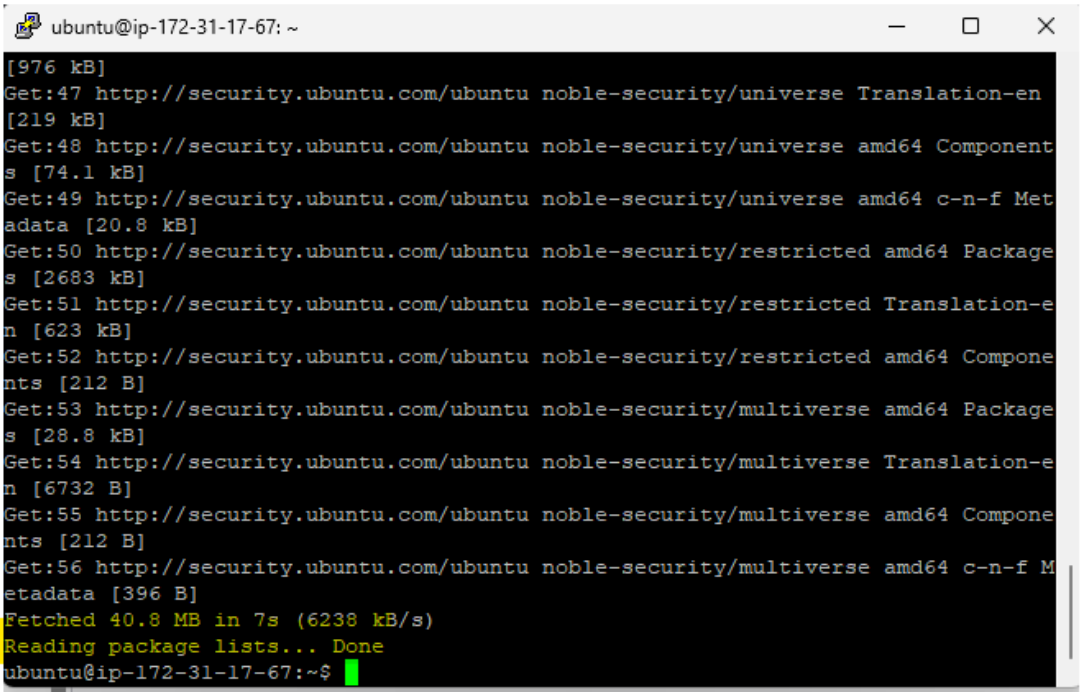


Figure 4 b: sudo apt-get update

7. **Next, describe in depth what the *update* and *upgrade* commands are doing (explain why it is important to run these commands, how often should these commands be run, where do the commands pull the updates).**

The sudo apt-get update refreshes the local package index by fetching the latest list of available packages from Ubuntu's official repositories. The sudo apt-get upgrade

installs those newer versions. It is important to run these commands to maintain the CIA triad and patch security vulnerabilities (NIST SP 800-125). Run them at least weekly or after new software. The commands pull updates from official Ubuntu mirrors like us.archive.ubuntu.com.

8. **After successfully deploying the Ubuntu Server operating system updates and upgrades, reconnect to the AWS EC2 console, and if needed login again via an SSH client (Note: SSH session inactivity timeout is short). Instructions to SSH are provided by Amazon on how to do so. When connected, run the following commands to obtain information about the host and network settings (perform a screen capture of the results and embed below):**

   echo '<your name>' && echo 'CMIT 495 <your section> <your semester>' && date *(section is a 4-digit number like 6380 and semester is a 4-digit number like 2215 or 2218)*

   whoami

   ip

   pwd

   ping -c 4 www.google.com

ubuntu@ip-172-31-17-67: ~

```
ubuntu@ip-172-31-17-67:~$ echo '<Aron Michaels>' && echo 'CMIT 495 7393 2262' &&
 date
<Aron Michaels>
CMIT 495 7393 2262
Wed Mar 18 02:16:52 UTC 2026
ubuntu@ip-172-31-17-67:~$ whoami
ubuntu
ubuntu@ip-172-31-17-67:~$ ip a show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group defa
ult qlen 1000
    link/ether 0a:ff:d8:dc:2b:33 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    inet 172.31.17.67/20 metric 100 brd 172.31.31.255 scope global dynamic ens5
       valid_lft 3581sec preferred_lft 3581sec
    inet6 fe80::8ff:d8ff:fedc:2b33/64 scope link
       valid_lft forever preferred_lft forever
ubuntu@ip-172-31-17-67:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-17-67:~$ ping -c 4 www.google.com
PING www.google.com (172.253.62.105) 56(84) bytes of data.
64 bytes from bc-in-f105.1e100.net (172.253.62.105): icmp_seq=1 ttl=106 time=1.48 ms
64 bytes from bc-in-f105.1e100.net (172.253.62.105): icmp_seq=2 ttl=106 time=1.51 ms
64 bytes from bc-in-f105.1e100.net (172.253.62.105): icmp_seq=3 ttl=106 time=1.53 ms
64 bytes from bc-in-f105.1e100.net (172.253.62.105): icmp_seq=4 ttl=106 time=1.48 ms

--- www.google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.478/1.501/1.534/0.022 ms
ubuntu@ip-172-31-17-67:~$
```

Example Screenshot:

```
Windows PowerShell    linuxgeek@DILBERT: /mnt/c/Us

linuxgeek@DILBERT:/mnt/c/Users/g33k2$ echo 'John Galliano' && echo 'CMIT 495 6380 2215' && date
John Galliano
CMIT 495 6380 2215
Sun Mar 28 11:50:13 EDT 2021
linuxgeek@DILBERT:/mnt/c/Users/g33k2$ whoami
linuxgeek
linuxgeek@DILBERT:/mnt/c/Users/g33k2$ ip a show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: bond0: <BROADCAST,MULTICAST,MASTER> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 42:ae:26:92:7c:2b brd ff:ff:ff:ff:ff:ff
3: dummy0: <BROADCAST,NOARP> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether a6:d3:51:b3:f4:76 brd ff:ff:ff:ff:ff:ff
4: tunl0@NONE: <NOARP> mtu 1480 qdisc noop state DOWN group default qlen 1000
    link/ipip 0.0.0.0 brd 0.0.0.0
5: sit0@NONE: <NOARP> mtu 1480 qdisc noop state DOWN group default qlen 1000
    link/sit 0.0.0.0 brd 0.0.0.0
6: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:05:a2:b7 brd ff:ff:ff:ff:ff:ff
    inet 172.24.167.20/20 brd 172.24.175.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe05:a2b7/64 scope link
       valid_lft forever preferred_lft forever
linuxgeek@DILBERT:/mnt/c/Users/g33k2$ pwd
/mnt/c/Users/g33k2
linuxgeek@DILBERT:/mnt/c/Users/g33k2$ ping www.google.com
PING www.google.com (142.250.64.68) 56(84) bytes of data.
64 bytes from lga34s30-in-f4.1e100.net (142.250.64.68): icmp_seq=1 ttl=118 time=15.6 ms
64 bytes from lga34s30-in-f4.1e100.net (142.250.64.68): icmp_seq=2 ttl=118 time=14.9 ms
64 bytes from lga34s30-in-f4.1e100.net (142.250.64.68): icmp_seq=3 ttl=118 time=14.7 ms
^C
--- www.google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 14.749/15.076/15.592/0.368 ms
linuxgeek@DILBERT:/mnt/c/Users/g33k2$
```

9. **What kind of account did you discover when you ran the *whoami* command? How do you know?**

10. **Note the difference between IP addresses—specifically, when you perform the *ip a show* or *ipconfig* on your personal system, versus the *ip a show* command within the EC2 instance VM command prompt. Describe the network settings of each system. Why are the IP addresses different? Are the IP addresses private or public? What is the difference between a public and private IP address? Explain in detail.**

On my local Windows system, ipconfig showed a private IP address like 192.168.1.x (from my home router's LAN, with subnet mask 255.255.255.0 and default

gateway). The network is a typical home setup that is internal and not directly accessible from the internet.

In the VM, ip a show showed a private IP like 172.31.x.x (VPC internal) and a public IP like 3.x.x.x (assigned by AWS for external access). The VM's network is cloud-based with eth0 interface for AWS VPC.

The IPs are different because my local system is on a private home network while the VM is in AWS's cloud infrastructure. Local is private (not routable online); VM has both private (internal VPC) and public (routable). Public IPs are internet-facing for access (e.g., SSH) but riskier without firewalls; private IPs are for internal communication and safer for isolation (Learning Resource on virtual networking; NIST SP 800-125B on secure configs). In detail, public IPs (e.g., 3.0.0.0/8 range) allow global routing but need security groups; private (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) are non-routable, reducing exposure in virtual environments.

```
   Link-local IPv6 Address . . . . . : fe80::ed1e:ce2c:139e:189e%11
   IPv4 Address. . . . . . . . . . . : 192.168.116.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

thernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::1cbd:4b53:5d55:f819%21
   IPv4 Address. . . . . . . . . . . : 192.168.187.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

ireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : mynetworksettings.com
   IPv6 Address. . . . . . . . . . . : 2600:4040:b804:ed00:d19e:bc51:42c1:7c42
   Temporary IPv6 Address. . . . . . : 2600:4040:b804:ed00:45d8:90f7:d49b:9e1f
   Link-local IPv6 Address . . . . . : fe80::5280:1ec5:5c12:5371%19
   IPv4 Address. . . . . . . . . . . : 192.168.1.158
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::baf8:53ff:fe59:a22%19
                                       192.168.1.1
```

Figure 10: Local ipconfig Output

ubuntu@ip-172-31-17-67: ~

```
ubuntu@ip-172-31-17-67:~$ echo '<Aron Michaels>' && echo 'CMIT 495 7393 2262' &&
 date
<Aron Michaels>
CMIT 495 7393 2262
Wed Mar 18 02:16:52 UTC 2026
ubuntu@ip-172-31-17-67:~$ whoami
ubuntu
ubuntu@ip-172-31-17-67:~$ ip a show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group defa
ult qlen 1000
    link/ether 0a:ff:d8:dc:2b:33 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    inet 172.31.17.67/20 metric 100 brd 172.31.31.255 scope global dynamic ens5
       valid_lft 3581sec preferred_lft 3581sec
    inet6 fe80::8ff:d8ff:fedc:2b33/64 scope link
       valid_lft forever preferred_lft forever
ubuntu@ip-172-31-17-67:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-17-67:~$ ping -c 4 www.google.com
PING www.google.com (172.253.62.105) 56(84) bytes of data.
64 bytes from bc-in-f105.1e100.net (172.253.62.105): icmp_seq=1 ttl=106 time=1.48 ms
64 bytes from bc-in-f105.1e100.net (172.253.62.105): icmp_seq=2 ttl=106 time=1.51 ms
64 bytes from bc-in-f105.1e100.net (172.253.62.105): icmp_seq=3 ttl=106 time=1.53 ms
64 bytes from bc-in-f105.1e100.net (172.253.62.105): icmp_seq=4 ttl=106 time=1.48 ms

--- www.google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.478/1.501/1.534/0.022 ms
ubuntu@ip-172-31-17-67:~$
```

Figure 11: VM ip a show Output

11. Virtualization allows us to place the functionality of many servers into a single host while maintaining configurable levels of separation between all virtual machines. How do you believe virtualization could help in data center consolidation? What would you be leery about during and after any transition?

Virtualization helps in data center consolidation by running multiple VMs on a single physical host, which reduces the number of servers needed, lowers power and hardware costs, and simplifies management for Caduceus Inc. (Learning Resource on resource sharing). For example, instead of separate physical servers for different apps, one host can isolate them virtually, saving space and energy.

I would be leery about potential downtime during migration that could disrupt telehealth services mitigate with testing and phased rollout. After transition, watch for "noisy neighbor" issues where one VM hogs resources or VM escape vulnerabilities (NIST SP 800-125 on risks) mitigate with monitoring tools and regular patches.

12. Does virtualization increase the cybersecurity posture of the organization? If so, describe how and why. If not, describe how and why not.

Yes, virtualization increases the organization's cybersecurity posture by enabling better isolation, where threats in one VM don't spread to others (Learning Resource on separation). How: Features like snapshots for quick recovery enhance availability, and hypervisor controls limit access (NIST SP 800-125A on secure deployment). Why: It is more secure than physical setups for segmentation and reduces the attack surface in cloud environments like Caduceus's, but only if configured properly to avoid shared resource risks.

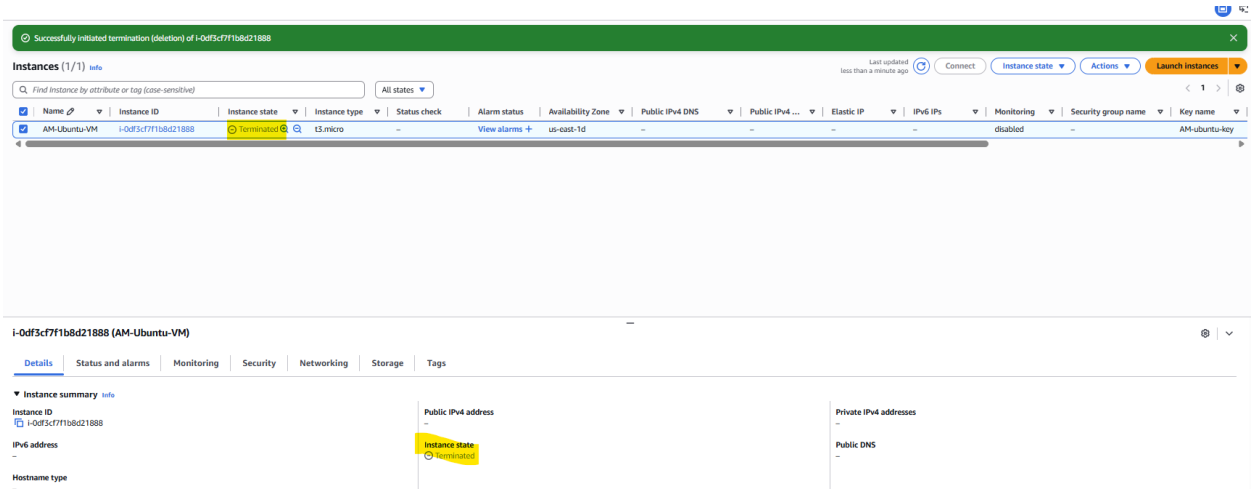
13. There are various virtualization options: bare-metal (Type 1) in which the hypervisors run directly on the hardware as their own operating systems, and user-space (Type 2) hypervisors that run within the conventional operating systems. Which of these options is more secure? Describe the vulnerabilities you believe exist in either Type 1, Type 2, or both configurations. What do you believe can be done to mitigate these vulnerabilities?

Type 1 (bare-metal) is more secure because it runs directly on hardware with a smaller attack surface and no host OS to exploit (NIST SP 800-125A on hypervisor deployment). Type 2 is less secure as attacks on the host OS can compromise all VMs.

Vulnerabilities in Type 1: Hypervisor bugs like VM escape (malware jumping to host). Type 2: Host OS flaws spreading to guests. Both: Resource sharing leading to side-channel attacks.

Mitigation: Regular patches, least privilege principles, monitoring logs, and segmentation (NIST SP 800-125 on best practices; Learning Resource on isolation).

14. Confirm that you have <u>stopped</u> and <u>terminated</u> your AWS Linux server instance. To confirm, simply type your name below.



termination

Aron Michaels

Export/print your completed file as a PDF Qand upload to your LEO/Assignments folder.